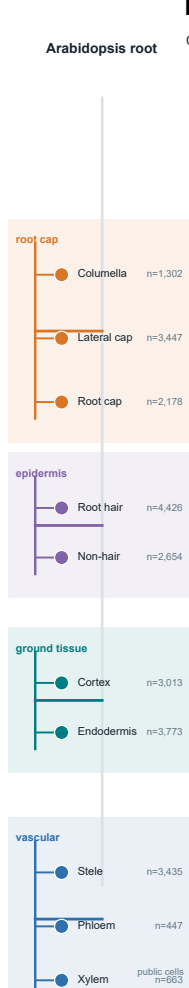
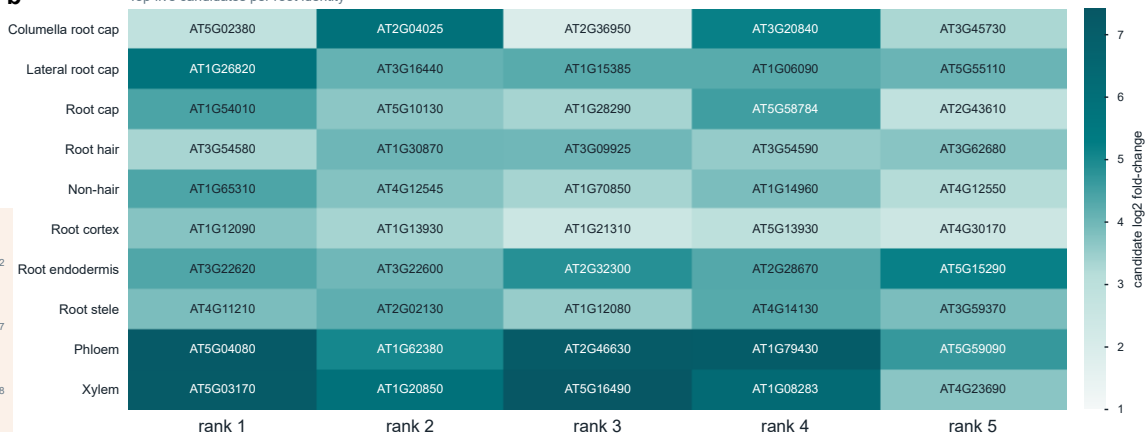


a Root taxonomy and cell scale

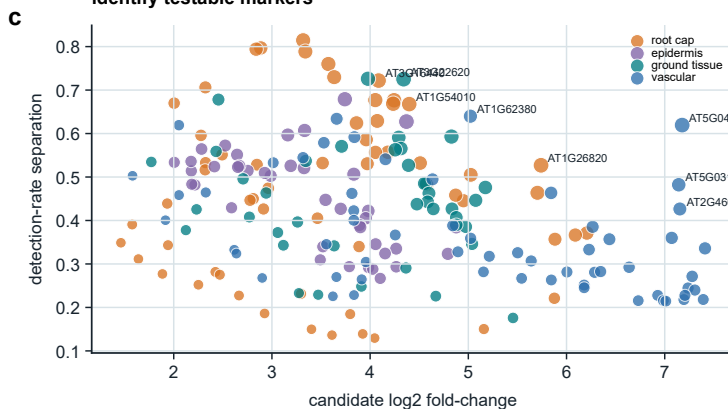


b State-resolved marker-candidate programs

Top five candidates per root identity



c Candidate effect size and detection separation identify testable markers



d Program strength differs across root identities

